



Homework 10: We will calculate the expectation value of the distance between two identical particles.

Consider the two-distinguishable-particle wavefunction

$$\psi_{\alpha\beta}(\mathbf{x}_1, \mathbf{x}_2) = (\langle \mathbf{x}_1 | \otimes \langle \mathbf{x}_2 |) |\psi\rangle_{12} = \psi_{\alpha}(\mathbf{x}_1) \psi_{\beta}(\mathbf{x}_2) \quad (1)$$

where $|\psi\rangle_{12} = |\alpha\rangle \otimes |\beta\rangle$ with $|\alpha\rangle$ and $|\beta\rangle$ single-particle states with arbitrary quantum number labels. For identical particles, we symmetrize (bosons) or anti-symmetrize (fermions):

$$\psi_{\alpha\beta}^{\pm}(\mathbf{x}_1, \mathbf{x}_2) = (\langle \mathbf{x}_1 | \otimes \langle \mathbf{x}_2 |) |\psi\rangle_{12}^{\pm} = \frac{1}{\sqrt{2}} [\psi_{\alpha}(\mathbf{x}_1) \psi_{\beta}(\mathbf{x}_2) \pm \psi_{\beta}(\mathbf{x}_1) \psi_{\alpha}(\mathbf{x}_2)] \quad (2)$$

where $|\psi\rangle_{12}^{\pm} = \frac{1}{\sqrt{2}} (|\alpha\rangle |\beta\rangle \pm |\beta\rangle |\alpha\rangle)$.

Question 1: calculate the expectation value of the distance between particles:

$$\langle (\mathbf{x}_1 - \mathbf{x}_2)^2 \rangle \quad (3)$$

You can assume that $\langle \alpha | \beta \rangle = 0$ and $\langle \alpha | \alpha \rangle = \langle \beta | \beta \rangle = 1$.

Bonus Question: do the same, but without assuming orthogonality of $|\alpha\rangle$ and $|\beta\rangle$. That is, take $\langle \alpha | \beta \rangle = S$ with $S \in \mathbb{C}$ and $|S| \leq 1$.



Answer 1: let me solve keeping S general already. The case $S \rightarrow 0$ then follows. Note that in that case, a canonical normalization of our state requires some tweaking. For distinguishable particles, it is trivially true that

$$\langle \psi |_{12} | \psi \rangle_{12} = \langle \alpha | \otimes \langle \beta | \alpha \rangle \otimes | \beta \rangle = \langle \alpha | \alpha \rangle \langle \beta | \beta \rangle = 1 \quad (4)$$

but for identical particles, we have

$$\langle \psi |_{12}^{\pm} | \psi \rangle_{12}^{\pm} = \frac{1}{2} (\langle \alpha | \alpha \rangle \langle \beta | \beta \rangle + \langle \beta | \beta \rangle \langle \alpha | \alpha \rangle \pm 2 \langle \alpha | \beta \rangle \langle \beta | \alpha \rangle) = 1 \pm |S|^2 \quad (5)$$

so we need to normalize our state as

$$|\psi\rangle_{12}^{\pm} = \frac{1}{\sqrt{2(1 \pm |S|^2)}} (|\alpha\rangle |\beta\rangle \pm |\beta\rangle |\alpha\rangle)$$

. Note that for $S = 1$ (i.e. $|\alpha\rangle = |\beta\rangle$), the fermionic state is not normalizable, as expected from the Pauli exclusion principle. For $S = 0$ we recover the stated normalization in the question.

we have

$$\langle (\mathbf{x}_1 - \mathbf{x}_2)^2 \rangle = \langle \mathbf{x}_1^2 \rangle + \langle \mathbf{x}_2^2 \rangle - 2 \langle \mathbf{x}_1 \cdot \mathbf{x}_2 \rangle \quad (6)$$

where the expectation values are taken with respect to different states in the three cases of interest.

Distinguishable particles: we start with x_1^2 . This is just the position operator in the first Hilbert space tensored with the identity for the second particle. Thus,

$$\langle \mathbf{x}_1^2 \rangle = (\langle \alpha | \otimes \langle \beta |) (\mathbf{x}_1^2 \otimes 1) (|\alpha\rangle \otimes |\beta\rangle) = \langle \alpha | \mathbf{x}^2 | \alpha \rangle \langle \beta | \beta \rangle = \langle \mathbf{x}^2 \rangle_{\alpha} \quad (7)$$

where $\langle \beta | \beta \rangle = 1$. Similarly for $\langle \mathbf{x}_2^2 \rangle = \langle \mathbf{x}^2 \rangle_{\beta}$. Note that the label on $\mathbf{x}_1$ was just to indicate the first particle, so we have dropped it when evaluating the expectation value of it in the single-particle Hilbert space.

Finally, we are left with the cross term:

$$\langle \mathbf{x}_1 \cdot \mathbf{x}_2 \rangle = (\langle \alpha | \otimes \langle \beta |) (\mathbf{x}_1 \otimes \mathbf{x}_2) (|\alpha\rangle \otimes |\beta\rangle) = \langle \alpha | \mathbf{x} | \alpha \rangle \langle \beta | \mathbf{x} | \beta \rangle = \langle \mathbf{x} \rangle_{\alpha} \cdot \langle \mathbf{x} \rangle_{\beta} \quad (8)$$

Overall,

$$\langle (\mathbf{x}_1 - \mathbf{x}_2)^2 \rangle = \langle \mathbf{x}^2 \rangle_{\alpha} + \langle \mathbf{x}^2 \rangle_{\beta} - 2 \langle \mathbf{x} \rangle_{\alpha} \cdot \langle \mathbf{x} \rangle_{\beta}. \quad (9)$$

In words: the expectation value of the distance can be evaluated as we would in a classical system, by treating the expectation values of the positions as classical variables.

Identical bosons: Now things get more interesting thanks to the multiple cross terms:

$$\begin{aligned} \langle \mathbf{x}_1^2 \rangle_{\pm} &= \left(\frac{\langle \alpha | \otimes \langle \beta | \pm \langle \beta | \otimes \langle \alpha |}{\sqrt{2(1 \pm |S|^2)}} \right) (\mathbf{x}_1^2 \otimes 1) \left(\frac{|\alpha\rangle \otimes |\beta\rangle \pm |\beta\rangle \otimes |\alpha\rangle}{\sqrt{2(1 \pm |S|^2)}} \right) \\ &= \frac{1}{2(1 \pm |S|^2)} (\langle \alpha | \mathbf{x}^2 | \alpha \rangle \langle \beta | \beta \rangle + \langle \beta | \mathbf{x}^2 | \beta \rangle \langle \alpha | \alpha \rangle \pm \langle \alpha | \mathbf{x}^2 | \beta \rangle \langle \beta | \alpha \rangle \pm \langle \beta | \mathbf{x}^2 | \alpha \rangle \langle \alpha | \beta \rangle) \\ &= \frac{1}{2(1 \pm |S|^2)} (\langle \mathbf{x}^2 \rangle_{\alpha} + \langle \mathbf{x}^2 \rangle_{\beta} \pm \langle \alpha | \mathbf{x}^2 | \beta \rangle S^* \pm \langle \beta | \mathbf{x}^2 | \alpha \rangle S) \\ &= \frac{1}{2(1 \pm |S|^2)} (\langle \mathbf{x}^2 \rangle_{\alpha} + \langle \mathbf{x}^2 \rangle_{\beta} \pm 2\Re(\langle \mathbf{x}^2 \rangle_{\alpha\beta} S^*)) \end{aligned} \quad (10)$$

where we used $\langle \beta | \alpha \rangle = S^*$ and defined $\langle \mathbf{x}^2 \rangle_{\alpha\beta} = \langle \alpha | \mathbf{x}^2 | \beta \rangle$. Similarly for $\langle \mathbf{x}_2^2 \rangle_{\pm}$.

Finally, the cross term is

$$\begin{aligned}
\langle \mathbf{x}_1 \cdot \mathbf{x}_2 \rangle_{\pm} &= \left(\frac{\langle \alpha | \otimes \langle \beta | \pm \langle \beta | \otimes \langle \alpha |}{\sqrt{2(1 \pm |S|^2)}} \right) (\mathbf{x}_1 \otimes \mathbf{x}_2) \left(\frac{|\alpha\rangle \otimes |\beta\rangle \pm |\beta\rangle \otimes |\alpha\rangle}{\sqrt{2(1 \pm |S|^2)}} \right) \\
&= \frac{1}{2(1 \pm |S|^2)} (\langle \alpha | \mathbf{x} | \alpha \rangle \langle \beta | \mathbf{x} | \beta \rangle + \langle \beta | \mathbf{x} | \beta \rangle \langle \alpha | \mathbf{x} | \alpha \rangle \pm 2 \langle \alpha | \mathbf{x} | \beta \rangle \langle \beta | \mathbf{x} | \alpha \rangle) \\
&= \frac{1}{1 \pm |S|^2} (\langle \mathbf{x} \rangle_{\alpha} \cdot \langle \mathbf{x} \rangle_{\beta} \pm \langle \alpha | \mathbf{x} | \beta \rangle \cdot \langle \beta | \mathbf{x} | \alpha \rangle) \\
&= \frac{1}{1 \pm |S|^2} (\langle \mathbf{x} \rangle_{\alpha} \cdot \langle \mathbf{x} \rangle_{\beta} \pm |\langle \alpha | \mathbf{x} | \beta \rangle|^2)
\end{aligned} \tag{11}$$

Overall, we have,

$$\begin{aligned}
(1 \pm |S|^2) \langle (\mathbf{x}_1 - \mathbf{x}_2)^2 \rangle_{\pm} &= \langle \mathbf{x}_1^2 \rangle_{\pm} + \langle \mathbf{x}_2^2 \rangle_{\pm} - 2 \langle \mathbf{x}_1 \cdot \mathbf{x}_2 \rangle_{\pm} \\
&= \langle \mathbf{x}^2 \rangle_{\alpha} + \langle \mathbf{x}^2 \rangle_{\beta} - 2 \langle \mathbf{x} \rangle_{\alpha} \cdot \langle \mathbf{x} \rangle_{\beta} \pm 2 \Re \left(\langle \mathbf{x}^2 \rangle_{\alpha\beta} S^* \right) \mp 2 |\langle \mathbf{x} \rangle_{\alpha\beta}|^2.
\end{aligned} \tag{12}$$

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Let us compare the difference between the distinguishable and identical particle cases for $S = 0$. First for the Fermions ($-$), we have:

$$\langle (\mathbf{x}_1 - \mathbf{x}_2)^2 \rangle_{-} - \langle (\mathbf{x}_1 - \mathbf{x}_2)^2 \rangle = +2 |\langle \mathbf{x} \rangle_{\alpha\beta}|^2. \tag{13}$$

This is positive. So the distance between two identical fermions is larger than the distance between two distinguishable particles (unsurprisingly).

Now, for bosons ($+$), we have

$$\langle (\mathbf{x}_1 - \mathbf{x}_2)^2 \rangle_{+} - \langle (\mathbf{x}_1 - \mathbf{x}_2)^2 \rangle = -2 |\langle \mathbf{x} \rangle_{\alpha\beta}|^2. \tag{14}$$

The fact that this distance wants to be smaller than the distinguishable case is behind the **exchange force** that makes bosons want to be close to each other.

A few further remarks on the general $S \neq 0$ case are in order.

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Limit checks. Setting $S = 0$ recovers the result of Question 1, as expected. For identical bosons with $S \rightarrow 1$ (i.e. $|\alpha\rangle \rightarrow |\beta\rangle$), the denominator $1 + |S|^2 \rightarrow 2$, and the matrix elements reduce to $\langle \mathbf{x}^2 \rangle_{\alpha\beta} \rightarrow \langle \mathbf{x}^2 \rangle_{\alpha}$, $|\langle \mathbf{x} \rangle_{\alpha\beta}|^2 \rightarrow |\langle \mathbf{x} \rangle_{\alpha}|^2$, and $\langle \mathbf{x} \rangle_{\alpha} \rightarrow \langle \mathbf{x} \rangle_{\beta}$. Therefore,

$$\langle (\mathbf{x}_1 - \mathbf{x}_2)^2 \rangle_{+} = 2 (\langle \mathbf{x}^2 \rangle_{\alpha} - |\langle \mathbf{x} \rangle_{\alpha}|^2) = 2(\Delta x)_{\alpha}^2. \tag{15}$$

Here, $(\Delta x)_{\alpha}^2$ is the variance of the position in the state $|\alpha\rangle$ and a measure of the spatial extent of the single-particle wavefunction. This makes intuitive sense: if both bosons are in the same state $|\alpha\rangle$, then their relative distance is determined by the spread of that state.